MILITARY SPECIFICATION

PRIMER, PERCUSSION, M100
METAL PARTS FOR

This specification is approved for use by U.S. Army Armament Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the metal parts for one type of percussion primer designated at Primer, Percussion, M100.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components

FSC: 1390

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DRAWINGS

ARMY

8838012	-	Cup, Battery
8838015	-	Plug, Firing
9327403	-	Plug, Closing
9277150	-	Head Primer

(Copies of specifications, standard, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Metal parts.-The metal parts shall comply with all requirements specified on Drawings (dwg.) 8838012, 8838015, 9327403 and 9277150, as applicable, and with all requirements specified in applicable specifications.

3.3 Workmanship.-All parts shall be fabricated and finished in a thorough, workmanlike manner. They shall be clean and free of burrs, sharp edges, unblended radii, surface defects, cracks, chips, dirt, grease, rust and other foreign matter. The cleaning method used shall not be injurious to any parts nor shall the parts be contaminated by the cleaning agent. All required marking shall be neat and sharply defined.

3.4 First article inspection.-This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance.-Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

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4.2 Classification of inspection.-The following types of inspection shall be conducted on this item:

a. First Article Inspection.

b. Quality Conformance Inspection.

4.3 First article inspection

4.3.1 Submission.-The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of 25 each of every component which have been produced by the contractor (or furnished to him by a supplier).

4.3.2 Inspection to be performed.-The first article parts shall be submitted in accordance with paragraph 4.3.1, and will be subjected by the Government to any or all the requirements of the applicable drawings and to the inspection in 4.4.

4.3.3 Rejection.-See MIL-A-48078.

4.4 Quality conformance inspection

4.4.1 Inspection lot formation.-Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination.-See MIL-A-48078.

a. Sampling Plans.-Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II, except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Cup, Battery		8838012
			NEXT HIGHER ASSEMBLY 9277103

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101.	Outside diameter, minimum (min)		.40%	3.2	Gage
102.	Thickness through bottom, maximum (max)		.40%	3.2	Gage
103.	Length		.40%	3.2	Gage
Minor					
201.	Diameter of hole		.65%	3.2	Gage
202.	Thickness through bottom, min.		.65%	3.2	Gage
203.	Inside diameter		.65%	3.2	Gage
204.	Outside diameter, max.		.65%	3.2	Gage
205.	Flat missing from surface at open end		.65%	3.2	Visual
206.	Surface finish improper		.65%	3.2	Visual
207.	Evidence of poor workmanship		.65%	3.3	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63215 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Plug, Firing				8838015
					NEXT HIGHER ASSEMBLY
					9277103
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101.	Large diameter, min.		.40%	3.2	Gage
102.	Length and profile of point		.40%	3.2	Gage
103.	Length of pilot, min.		.40%	3.2	Gage
104.	Total length		.40%	3.2	Gage
Minor					
201.	Large diameter, max.		.65%	3.2	Gage
202.	Length of pilot, max.		.65%	3.2	Gage
203.	Diameter of pilot		.65%	3.2	Gage
204.	Radii missing or improper		.65%	3.2	Visual
205.	Surface finish improper		.65%	3.2	Visual
206.	Evidence of poor workmanship		.65%	3.3	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63215 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
4.4.2.3	Head, Primer		9277150
			NEXT HIGHER ASSEMBLY 9277103

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1.	Crack or split		100%	3.2	Visual
2.	Flash hole missing		100%	3.2	Gage
Major					
101.	Pitch diameter of external thread, min.		.40%	3.2	Gage
102.	Major diameter of external thread, min.		.40%	3.2	Gage
103.	Pitch diameter of internal thread, max.		.40%	3.2	Gage
104.	Minor diameter of internal thread, max.		.40%	3.2	Gage
105.	Thickness of flange, max.		.40%	3.2	Gage
106.	Diameter of battery cup cavity		.40%	3.2	Gage
107.	Runout of battery cup cavity excessive		.40%	3.2	Gage
108.	Diameter of percussion element cavity		.40%	3.2	Gage
109.	Runout of percussion element cavity excessive			3.2	Gage
110.	Presence of flash hole at the bottom of the powder cavity		100%	3.2	Visual
Minor					
201.	Flange diameter		.65%	3.2	Gage
202.	Runout of flange diameter excessive		.65%	3.2	Gage
203.	Depth of battery cup cavity		.65%	3.2	Gage
204.	Depth to base of percussion element cavity		.65%	3.2	Gage
205.	Diameter of flash holes (5 holes)		.65%	3.2	Gage
206.	True position of flash holes excessive		.65%	3.2	Gage
207.	Distance from base to flash holes		.65%	3.2	Gage
208.	Diameter of external thread relief		.65%	3.2	Gage

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63215 (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.3 (con't)	Head, Primer				9277150
					NEXT HIGHER ASSEMBLY
					9277103
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
209.	Depth of slot(s)		.65%	3.2	Gage
210.	Width of slot(s)		.65%	3.2	Gage
211.	Total length		.65%	3.2	Gage
212.	Length from base to end of external threads		.65%	3.2	Gage
213.	Length from base to groove		.65%	3.2	Gage
214.	Length from flange to external threads		.65%	3.2	Gage
215.	Distance between slots		.65%	3.2	Gage
216.	Diameter of groove		.65%	3.2	Gage
217.	Distance from base to powder cavity		.65%	3.2	Gage
218.	Diameter of hole at bottom of percussion element cavity		.65%	3.2	Gage
219.	Diameter of powder cavity		.65%	3.2	Gage
220.	Outside diameter at top of primer head		.65%	3.2	Gage
221.	Distance from top of primer head to bottom of internal thread undercut		.65%	3.2	Gage
222.	Diameter of thread undercut, max.		.65%	3.2	Gage
223.	Effective thread, min.		.65%	3.2	Gage
224.	Chamfer(s) missing or improper		.65%	3.2	Visual
225.	Surface finish improper		.65%	3.2	Visual
226.	Protective finish missing or improper (if applicable)		.65%	3.2	Visual
227.	Radii missing or improper		.65%	3.2	Visual
228.	Evidence of poor workmanship		.65%	3.3	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.4	Plug , Closing		9327403
			NEXT HIGHER ASSEMBLY 9277103

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101.	Pitch diameter, min.		.40%	3.2	Gage
102.	Major diameter, min.		.40%	3.2	Gage
103.	Thickness		.40%	3.2	Gage
Minor					
201.	Width of slot		.65%	3.2	Gage
202.	Depth of slot (or radius, if applicable)		.65%	3.2	Gage
203.	Chamfer missing or incorrect		.65%	3.2	Visual
204.	Surface finish improper		.65%	3.2	Visual
205.	Protective finish missing or improper		.65%	3.2	Visual
206.	Evidence of poor workmanship		.65%	3.3	Visual
207.	**Pitch diameter, max.**		**.65%**	**3.2**	**Gage**
208.	**Major diameter, max.**		**.65%**	**3.2**	**Gage**

NOTES:

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4.4.3 Inspection equipment.-The inspection equipment required to perform the examinations and tests prescribed herein is described in the Paragraph Reference/Inspection Method column in tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078.

5. PREPARATION FOR DELIVERY

5.1 Packing ang marking

5.1.1 Level C.-Packing and marking shall be in accordance with MIL-STD-1169.

6. NOTES

6.1 Ordering data.-See MIL-A-48078.

6.2 Submission of designs for approval.-See 6.2.3 of MIL-A-48078. Submit equipment designs as required, to Commander, ARRADCOM, ATTN: DRDAR-QAR-I, Dover, New Jersey 07801.

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Preparing Activity:
Army-AR

Project Number: 1390-A169

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